KAVACH AI — Multilingual Voice Scam & Vishing Interception Platform

Great Indian AI Internship Challenge 2026 by Gnani AI • Track: Voice AI / Cybersecurity

1. The Crisis: India's Voice Social-Engineering Epidemic

Phone-based cyber fraud in India has evolved into an industrialized threat costing citizens thousands of crores annually. Criminal syndicates deploy high-pressure psychological extortion: **Digital Arrest rackets** (impersonating CBI/Police demanding illegal bail transfers), **OTP & Banking Credential Harvesting** (coercing immediate password reveals under threat of account freeze), and **Regional Linguistic Exploitation** (conversing in Hindi, Telugu, Hinglish, and Tenglish to bypass English-only filters).

2. The Solution: KAVACH AI Architecture

KAVACH AI provides an autonomous, real-time voice threat defense platform combining **Gnani AI speech intelligence** with a **cascaded Hybrid Intelligence Engine**:

Tier	Technology & Model	Core Function & Safety Invariant
Acoustic Ingestion	Gnani Prisma STT v3.0	Streams multilingual call audio; transcribes Hindi, Telugu, English with LID and timestamped segment accuracy.
Stage 0: Pre-ML Gate	Rule-Priority Shield	Deterministic safety barrier. Critical threats (Digital Arrest, OTP demands) are locked to SCAM (Risk ≥ 85); zero ML downgrades.
Stage 1: Fast Path	Calibrated TF-IDF v3	Sublinear n-gram classifier resolving ~68% of standard traffic in **4.29 ms** CPU latency (~42 MB RAM).
Stage 2: Deep Path	Google MuRIL v1 (237M)	Deep contextual transformer fine-tuned on Dataset v3. Selectively invoked for ambiguous/code-mixed calls, cutting compute by 68%.
Voice Response	Gnani Timbre TTS v2.5	Synthesizes immediate audible defensive warnings in victim's language (Hindi, Telugu, English), prompting immediate disconnection.
SOC & Forensics	ReportLab PDF Engine	Generates tamper-evident forensic PDF incident dossiers for reporting to the National Cybercrime Helpline (1930).

3. Key Achievements & Verification Metrics

- **100% Test Accuracy & 0 Scam False Negatives**: Evaluated on the held-out 450 test split of Dataset v3 across 5 linguistic splits.
- **68% Compute Overhead Reduction**: Validation threshold calibration (τ=0.85) selectively escalates only 32% of calls to MuRIL, preserving a median latency of 4.29 ms.
- **Production Baseline Preserved**: Original production classifier v1 and existing endpoints remain untouched and regression-free.
- **78/78 Tests Passed**: 100% pass rate across the full backend regression suite; frontend compiles with 0 errors.
- **CPU Inference Ready**: Fully functional on standard commodity multi-core consumer hardware without requiring expensive GPUs.